

Marine Conservation and the Roles Aquariums Play

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COMP 3125 – Data Science
Fundamentals

Abstract—This electronic document is a “live” template and already defines the components of your paper [title, text, heads, etc.] in its style sheet. ***CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.** (provide a short abstract)

Keywords—example1, example2, example3, example 4, example 5 (provide 3-5 keywords)

I. INTRODUCTION (HEADING 1)

Marine Conservation is a very topical discussion now, given the current concerns over global warming and climate change. Marine life is something that should be preserved and protected for both the benefits of them and us. At the forefront of this conservation are aquariums, which play a vital role in preserving endangered species and engaging the public more on marine conservation. Most aquariums are non-profit, meaning any money they make are sent towards conservation funds or maintenance. Some however, like SeaWorld, are for profit, and can hurt the conservation efforts of others. This report plans to see just how big of role aquariums play in marine conservation.

II. DATASETS

A. Source of dataset (Heading 2)

In this section, introduce your dataset by explaining its source—where you obtained it and whether it is from a credible provider. Include details such as when the dataset was generated and how it was created by its original author. If you generated the dataset yourself, describe the methods and processes you used.

<https://www.aza.org/zoo-and-aquarium-statistics>

The Association for Zoos and Aquariums, or AZA as it's better known, is one of the most credible sources for information pertaining to marine conservation and a comparison of for-profit and non-profit institutions. Many of the sets have very recent updates, the oldest being April 2023.

B. Character of the datasets

Describe the dataset's format and size. Additionally, provide an overview of the dataset's characteristics, including its features, size, structure, and any relevant attributes that are important for your analysis. Describe the dataset's format and size, as well as its key features, including the parameters, columns, rows, and character attributes along with their respective units. Using a table to present this information is recommended for clarity. Explain whether you cleaned the data or converted any units, specifying the formulas or rules applied. If multiple datasets were combined, describe how they were merged. Additionally, mention if you created any new categories for analysis, detailing what they are and how they were generated. Providing this background ensures transparency and helps readers understand the reliability and relevance of your data.

The dataset is clean, only needing some trimming for a more accurate appraisal. The most relevant attributes are for the conservation funds and for the non-profit/for-profit breakdown.

III. METHODOLOGY

In this part, you should give an introduction of the methods/model. First, what's the method/model. What's the assumption of this method/model. What's the advantage/disadvantage of this method/model. Why did you choose it. What Python module or function do you apply to apply this method/model. Any optional input/extra work did you adjust to make the results better. If you have multiple methods, feel free to use subsection A., B. to separate them.

The model I will be planning on using is a linear regression model, which attempts to model the relationship between a target variable and predictor variables via a linear equation:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots \beta_n X_n + \epsilon.$$

Linear regression assumes multiple things. It assumes that the relationship between the predictors and the target variable is a linear one, assumes that residuals are independent of one another, constant across all levels of the predictors, and roughly follow a normal distribution, and finally that the predictor variables are not extremely correlated with one another.

Linear regression has its advantages as well as its disadvantages. For one, it's easily accessible, as the coefficients show how each variable affects a prediction. It's fast to compute, and it's easy to visualize. However, it's also extremely sensitive to outliers, only assumes linear relationships, and can predict decimals or negatives. Despite this, linear regression was chosen for its ease of accessibility, and the dataset being decently small compared to others means the model will have an even easier time computing it.

The Python modules that will likely be used will be Pandas, a data frame library, and Seaborn as the main visualizer for linear regression. This is because Seaborn has built in regression plotting, is great for comparing multiple variables, and can quickly check linear assumptions.

IV. RESULTS

In this section, present your findings using an appropriate method, such as equations, numerical summaries, or visualizations like charts and graphs. Clearly explain all results and provide guidance on how to interpret them. If any unexpected results arise, discuss possible reasons or contributing factors. To improve clarity and organization, consider using subsections (e.g., A, B) to separate different aspects of your results.

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this newly created file, highlight all of the contents and import your prepared text file. You are now ready to style your paper; use the scroll down window on the left of the MS Word Formatting toolbar.

A. Result A

Example: XXX

1) For papers with more than six authors: Add author names horizontally, moving to a third row if needed for more than 8 authors.

2) For papers with less than six authors: To change the default, adjust the template as follows.

a) Selection: Highlight all author and affiliation lines.

b) Change number of columns: Select the Columns icon from the MS Word Standard toolbar and then select the correct number of columns from the selection palette.

c) Deletion: Delete the author and affiliation lines for the extra authors.

B. Results B

Example: Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

C. Results C

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE I. TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^a Sample of a Table footnote. (Table footnote)

Fig. 1. Example of a figure caption. (figure caption)

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

V. DISCUSSION

Every method/project has its shortage or weakness. Please discuss the unsatisfied results in your project. And discuss the

feasible suggestions of future work to revise/improve your result.

Example: xxx

VI. CONCLUSION

In this part, you should summarize your project. What important results did you find for your topic and what’s the effect of this result on the real-world?

Example: xxx

ACKNOWLEDGMENT (Heading 5)

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

Use the IEEE format for the citation. The template will number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first ...” Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

[1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955. (references)

[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

[4] K. Elissa, “Title of paper if known,” unpublished.

[5] R. Nicole, “Title of paper with only first word capitalized,” J. Name Stand. Abbrev., in press.

[6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].

[7] M. Young, The Technical Writer’s Handbook. Mill Valley, CA: University Science, 1989.

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